Tristan Visser

FINAL-YEAR BSC COMPUTER SCIENCE STUDENT · BACKEND AND FULL-STACK DEVELOPER

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PROFILE

Final-year Computer Science student at Stellenbosch University with a strong foundation in backend architectures, relational database design, REST APIs and Agile workflows. I've built a graph-based dependency engine with Electrum engineers, a reliable UDP file-transfer protocol, and database-backed web applications for commercial clients. Looking for a graduate software engineering role from 2027.

EDUCATION

2023 — 2026

BSc Computer Science

Stellenbosch University

Focus: backend architectures, relational databases, REST APIs, software engineering.

EXPERIENCE

Freelance

Commercial Web Developer & Technical Consultant

Local business clients

- Designed, developed and deployed custom web applications backed by relational databases and secure hosting.
- Managed requirements, domain and DNS configuration, responsive UI and ongoing maintenance.

2026

Dependency Analysis Framework

Electrum · industry project

Built a Java dependency graph engine that identifies breaking API changes and opens automated GitHub pull requests for affected repositories, across three Agile sprints.

SELECTED PROJECTS

2026

Electrum Dependency Analysis Framework

A graph-based dependency engine that detects breaking API changes across repositories and opens pull requests for affected dependents.

JAVA · REST APIS · GITHUB API · CI/CD · AGILE / SCRUM · SEMANTIC VERSIONING

Case study

2026

P2P Encrypted File Sharing

A peer-to-peer file-sharing desktop app with a custom reliable UDP protocol, TCP signalling and end-to-end encryption.

JAVA · JAVAFX · UDP · TCP · END-TO-END ENCRYPTION

Case study

2026

BPE DFA Tokenizer

A deterministic finite automata engine for BPE subword tokenization, built to parse the token dictionaries used by language models.

PYTHON · JAVA · DFA · REGULAR LANGUAGES

Case study

2026

Parallel Process Scheduler

A multi-threaded CPU scheduler in C and OpenMP with IPC mailboxes, graph-based cycle detection and deadlock prevention.

C · OPENMP · LINUX · MULTI-THREADING

Case study

SKILLS

LANGUAGES	Java · Python · JavaScript · SQL · C
BACKEND AND DATA	PostgreSQL · Supabase · REST APIs · Node.js · Express.js
FRONTEND	React.js · Next.js · Tailwind CSS · HTML · CSS · WordPress
SYSTEMS AND TOOLS	C/OpenMP · Multi-threading · UDP/TCP · Linux/Bash · Git · CI/CD · Cloudflare · Vercel

ELSEWHERE

GitHub LinkedIn Email

LAST UPDATED: OCTOBER 2026 · REFERENCES AVAILABLE ON REQUEST.